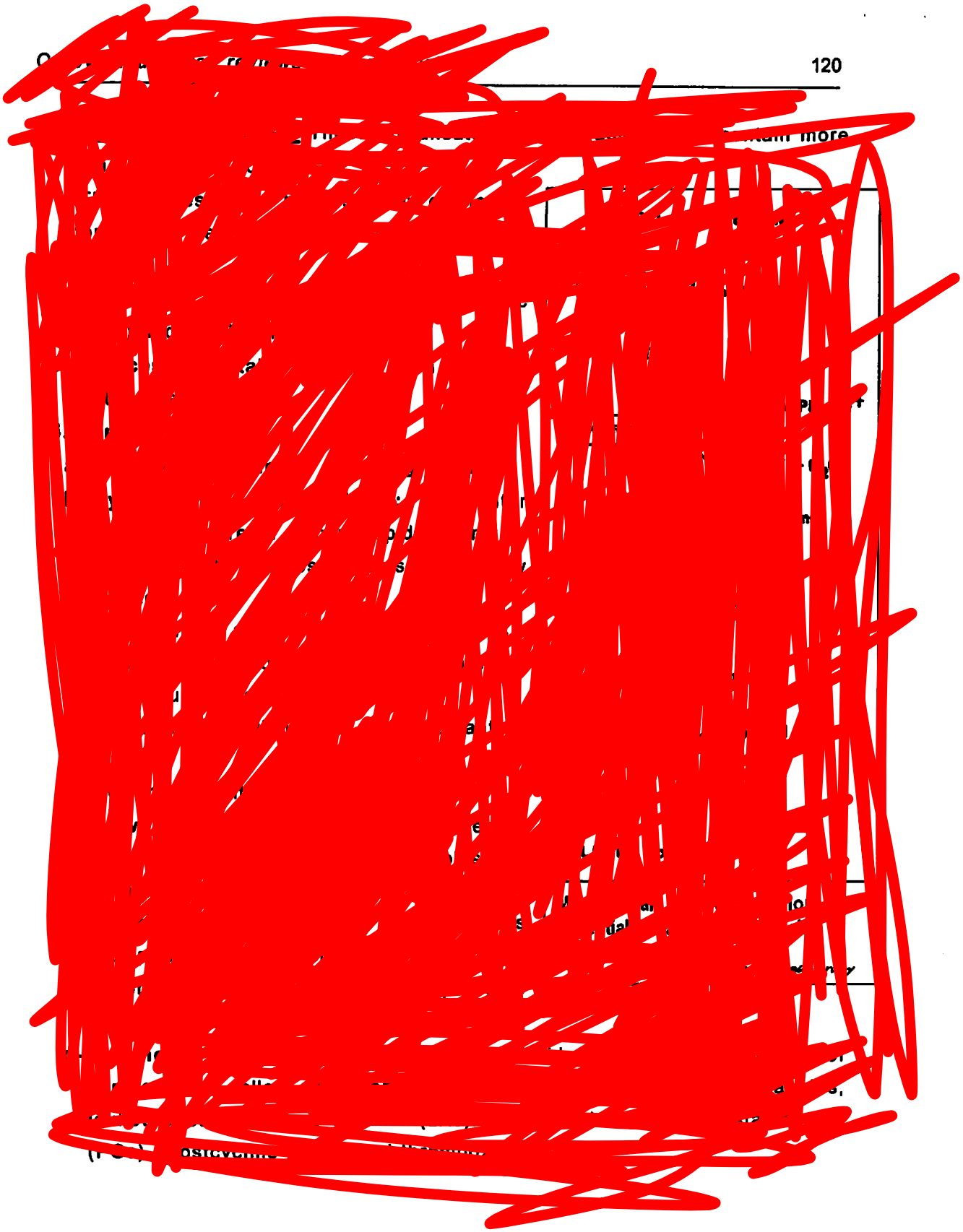
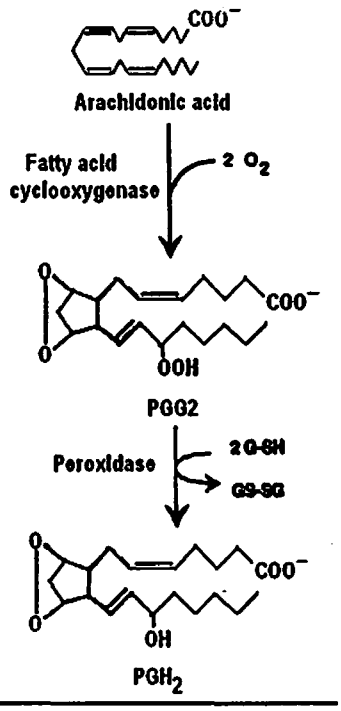


EICOSANOIDS

I. **Definition:** These are cyclic compounds that derived from arachidonic acid (eicosatetraenoic) (20 C) after cyclization of its carbons chain to form a ring.

II. Components of eicosanoids:

1. **Prostanoids:** which comprise prostaglandins, prostacyclins and thromboxanes. **Prostaglandins (PG) include many types** (A, B, D, E, F, H, G and I).
2. **Leukotriens (LT):**
   a) They are present in leucocytes, platelets and mast cells.



Oxidation and cyclization of arachidonic acid

III. *SYNTHESIS OF PROSTANOIDSS:*

A. The immediate precursor of prostanoids is ***arachidonic acid,*** which is 20 carbon polyunsaturated fatty acid that containing 4 double bonds (20:4 $\Delta^{5,8,11,14}$). Arachidonic acid is derived from **Phospholipids** present in all cell membranes by the action of phospholipase A_2 enzyme.

B. The first step in the synthesis of prostanoids is the oxidation and cyclization of arachidonic acid to give PGG_2 and PGH_2 (see the figure).

These reactions are catalyzed by two microsomal enzymes:

1. **Fatty acid cyclo-oxygenase**, which requires 2 oxygen molecules.
2. **Peroxidase,** which is dependant on reduced glutathione (G-SH).

C. PGH2 is the precursor all prostanoids.

D. Regulation of synthesis of prostanoids:

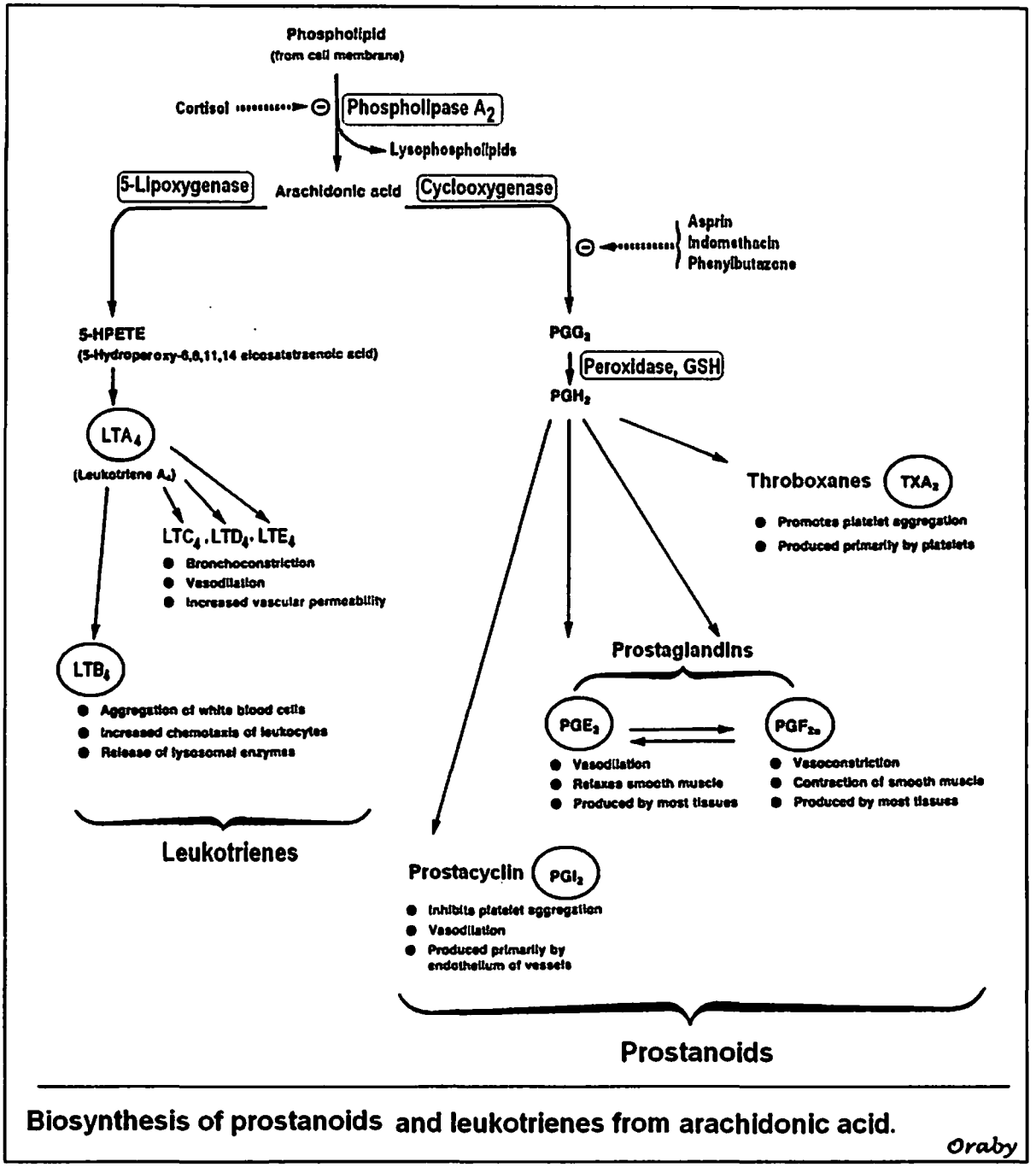
1. **Cortisol** inhibits the phospholipase A_2 activity,
2. **Aspirin,** indomethacin and phenylbutazone -which are antiinflamatory agents- inhibit cyclooxygenas. These agents do not affect synthesis of the leukotrienes.

IV. *SYNTHESIS OF LEUKOTRIENES:*

A. In neutrophils, arachidonic acid is converted to 5-hyroperoxy 6, 8, 11, 14 eicosatetraenoic acid (5-HPETE). This reaction is catalyzed by 5-lipooxygenase enzyme.

B. 5-HPETE is converted to a series of ***leukotrienes, (LT):*** LTA_4, LTB_4, LTC_4, LTD_4 and LTE_4.

C. In contrast to prostanoids synthesis, no drugs are known specifically to inhibit the lipoxygenase pathway.



Biosynthesis of prostanoids and leukotrienes from arachidonic acid.

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FUNCTIONS OF EICOSANOIDS:

A. ***Prostanoids:***

1. ***Thromboxanes:***
   - *a)* It is produced by platelets. It **stimulates platelets aggregations**.
2. ***Prostacyclin:***
   - *a)* **It inhibits platelets aggregations**.It is produced by endothelial cells of the blood vessels.
   - *b)* Vasodilatation.
3. ***Prostaglandins:***
   - *a)* **Prostaglandin E2 (PGE2):** Produced by most tissues.
     1) Vasodilatation.

2) Relaxesation of smooth muscles.

b) **Prostaglandin F2α (PGF2α):** Produced by most tissues.

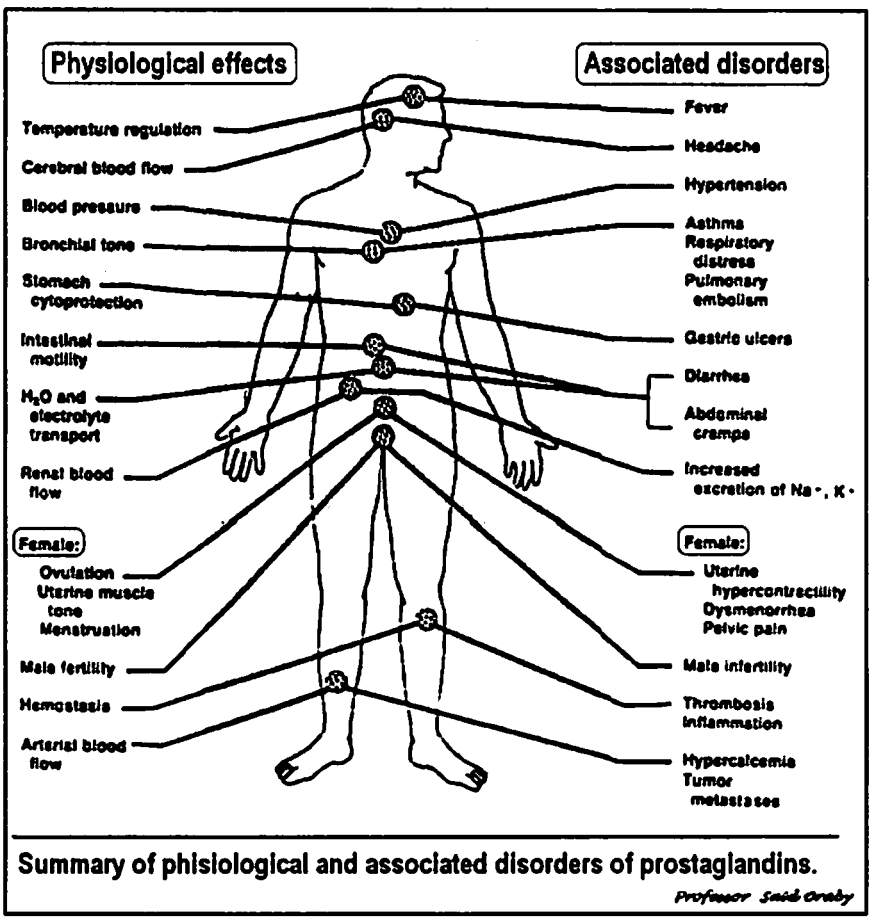
1) Vasoconstriction.

2) Contraction of smooth muscles.

> ***Note:*** Prostanoids have a **hormonal like action.** Prostanoidss differ from a true hormone in the following aspects:
>
> 1. They are formed in almost all tissues rather than in specialized glands.
> 2. They generally act locally rather than by circulating in the blood to the far target tissues.
> 3. Prostanoids are metabolized to inactive products at their site of synthesis and are not stored in any tissues.

B. ***Leukotrienes*** **(LTA_4):**

1. **Leukotrienes B_4 (LTB_4):**
   - *a)* **Movement and aggregation of white blood cells at the site of inflamation (chemotactic movement).**
   - *b)* **Release of lysosomal enzymes.**
2. **Leukotrienes (LTC_4, LTD_4, LTE_4):**
   - *a)* **Bronchoconstriction.**
   - *b)* **Vasodilatation.**
   - *c)* **Increased vascular permeability.**



Summary of phisiological and associated disorders of prostaglandins.

Professor Said Orabi

C. In addition to the above functions, ecosanoids have other wide range of physiological functions as regulation of body temperature and controlling blood pressure. These functions are shown in the above figure.

D. Excess production of prostanoids results in group of symptoms including pain, inflammation, fever, nausea and vomiting (see figure).

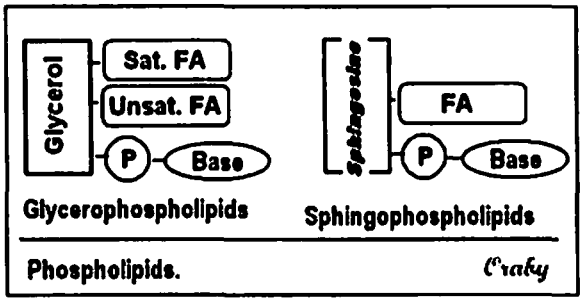
Metabolism of Conjugated Lipids

I. METABOLISM OF PHOSPHOLIPIDS:

A. Phospholipids are lipids containing phosphate.

B. They also contain alcohol (glycerol or sphingosine), fatty acid(s) and nitrogenous base.

C. The bases include; choline, serine, ethanolamine and inositol.



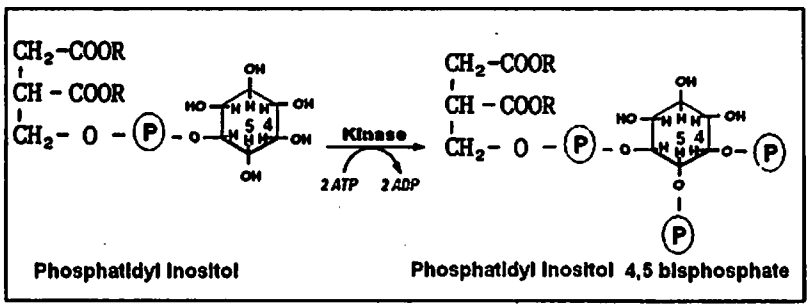
D. **Synthesis of lecithin (Phosphatidyl choline):** It is formed of glycerol, two fatty acids, phosphate and choline.

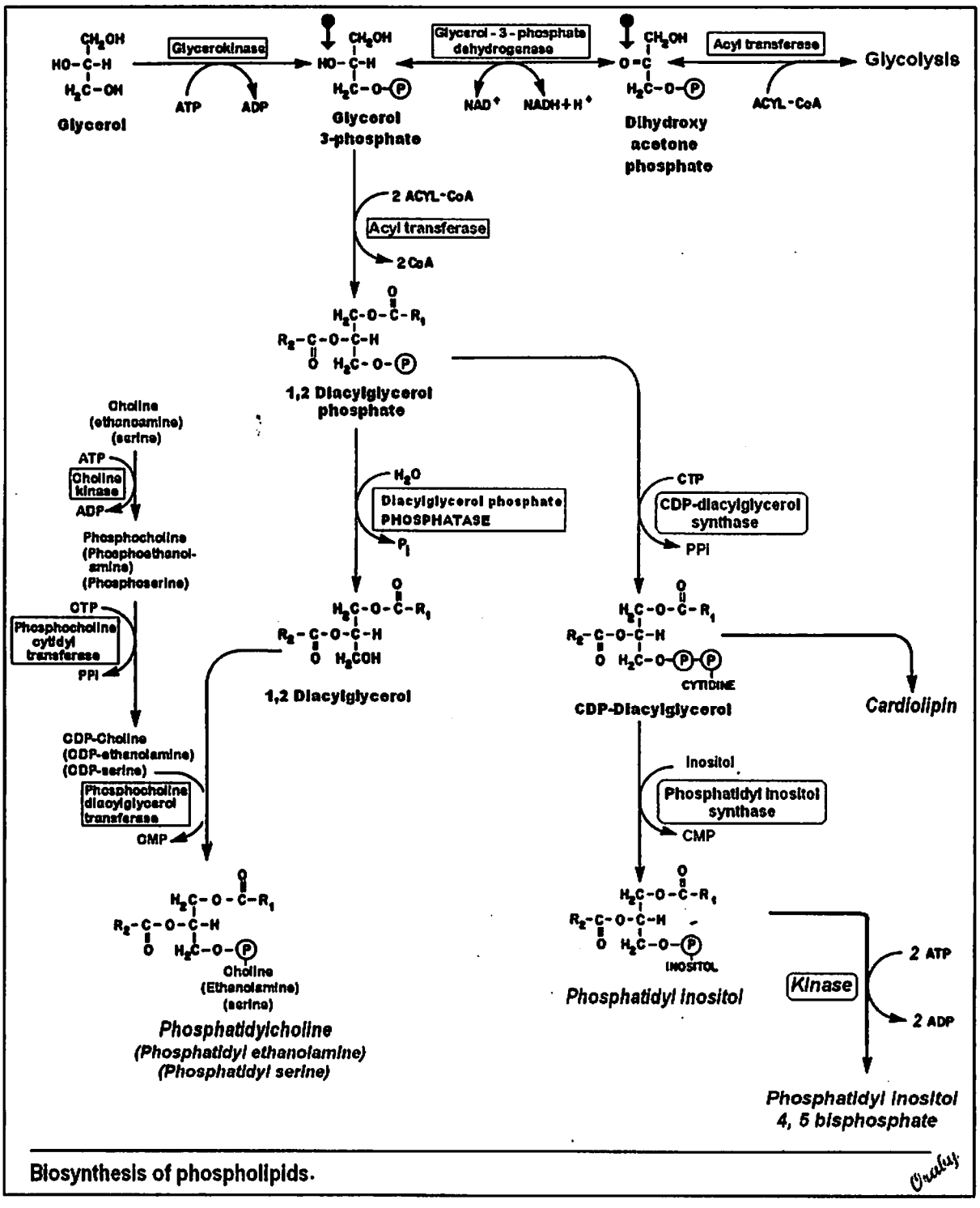
1. Activation of fatty acids into acyl CoA:
2. Synthesis of 1,2 diacylglycerol.
3. Activation of choline into CDP-choline.
4. Reaction of 1,2 diacylglycerol with CDP-choline to form lecithin.

E. **Synthesis of Phosphatidyl ethanolamine and phosphatidyl serine:** Same as lecithin but active ethanolamine (CDP-ethanolamine) or active serine (CDP-serine) are used instead of active choline.

F. **Synthesis of Phosphatidyl inositol and Phosphatidyl inositol 4,5 bisphosphate:**

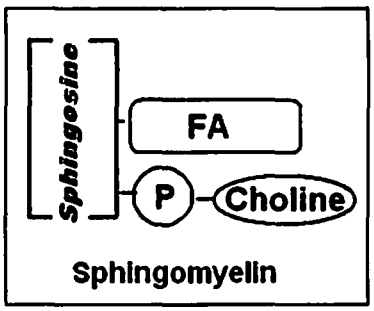
1. Activation of fatty acids into acyl CoA:
2. Synthesis of 1,2 diacylglycerol phosphate.
3. Reaction of 1,2 diacylglycerol phosphate with CTP to form CDP-diacylglycerol.
4. CDP-diacylglycerol reacts with inositol to form Phosphatidyl inositol.
5. Phosphatidyl inositol then accepts 2 phosphate from 2 ATP to form Phosphatidyl inositol 4, 5 bisphosphate.



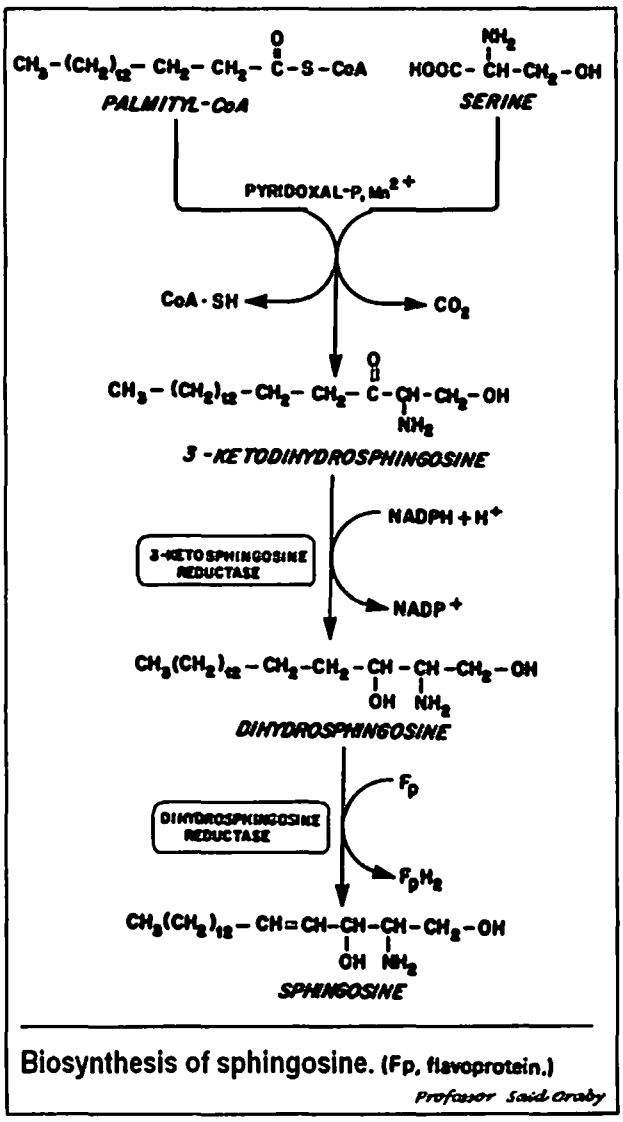


Biosynthesis of phospholipids.

G. **Synthesis of sphingomylin:** It is formed of: Sphingosine base, Fatty acyl CoA, phosphate and choline

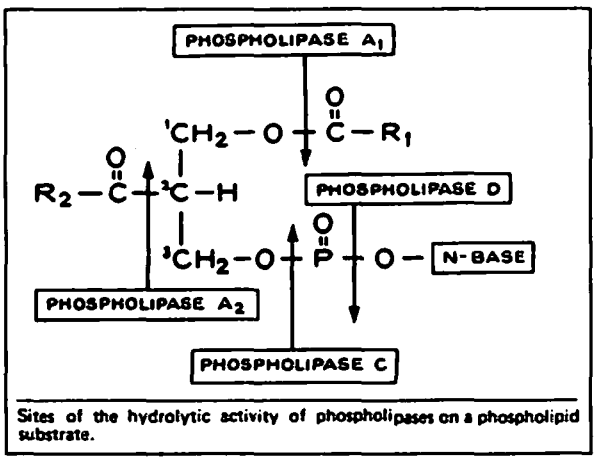


1. **Synthesis of sphingosine:** It is formed of amino acid serine and active palmitic acid (16C): palmityl CoA ➔ sphingosine.
2. Then sphingosine reacts with acyl CoA to form ceramide.
3. Ceramide then reacts with CDP-choline to form sphingomyelin.



Biosynthesis of sphingosine. (Fp, flavoprotein.)

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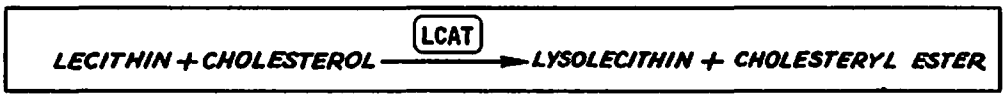


Sites of the hydrolytic activity of phospholipases on a phospholipid substrate.

H. **Degradation of phospholipids:**

1. Phosphoglycerides are degraded by phosphlipases A_1, A_2, C and D. They are present in most tissues and pancreatic juice.
2. Sphingomyelin is degraded by lysosomal enzymes, **sphingomyelinase. Its deficiency leads to a disease called** Niemann Pick disease. It is characterized by enlarged liver and spleen and death at early life.

3. A plasma enzyme called **LCAT** (lecithin cholesterol acyl transeferase) can act upon second fatty acyl CoA of lecithin (similar to phospholipase A2) converting it into lysolecithin.



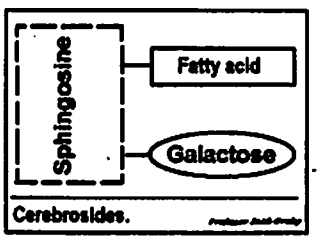
I. Functions of phospholipids:

1. Phospholipids enter in the **structure of cell membrane**.
2. Phospholipids containing choline (e.g. lecithin) act as **neurotransmitters**. They also act as **methyl donors** in transmethylation reaction.
3. Phospholipids act as **lipotropic factors** i.e. prevent accumulation of fat in liver and hence prevent fatty liver.
4. Dipalmityl lecithin acts as **surfactant** in the lungs. It prevents adherence of alveolar wall. Its deficiency leads to **respiratory distress** syndrome in premature babies.
5. Cephalin has a role in **coagulation** mechanism.
6. Lipositol (Phosphatidyl inositol) acts as precursor for **inositol triphosphate**. The latter acts as **2nd messenger**, mediating action of some hormones inside target cells.
7. Phospholipids in bile **make cholesterol soluble**. Their deficiency leads to cholesterol gallstones.

II. *METABOLISM OF GLYCOLIPIDS (SPHINGOLIPIDS):*

A. Glycolipids are lipids containing carbohydrates.

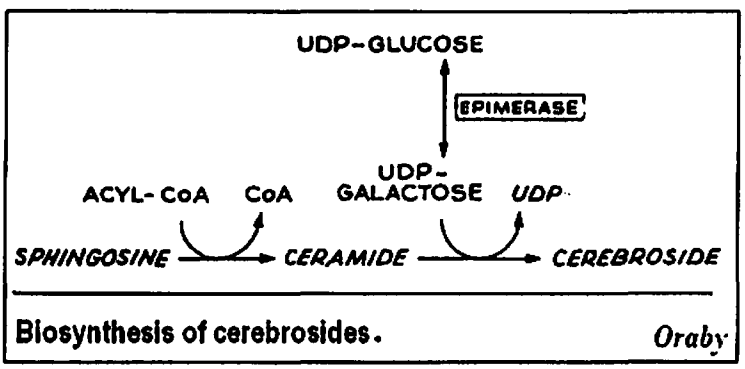
B. They also contain sphingosine and one fatty acid. They differ from each other according to the type of carbohydrate content.



C. **Synthesis of cerebrosides:** They Contain galactose.

1. Synthesis of sphingosine; See before.
2. Formation of ceramide.
3. Activation of galactose: ***UDP-Glucose → UDP Galactose.***

D. ***Degradation of cerebrosides*** is by ***glucocerebrosidase.*** Its defiency leads to a disease called **Gaucher's disease.** It is characterized by mental retardation and enlarged liver and spleen in children.



E. Lipidosis: Errors of phospholipids and sphingolipids metabolism:

These are a group of diseases in which there is an abnormal accumulation of phospholipids and glycolipids in nervous tissue. They are common in children. They are characterized by:

1. Deficiency of specific lysosomal enzymes responsible for degradation of sphingolipids.
2. Accumulation of sphingolipids in tissues leads to their enlargement e.g. liver and spleen enlargement.
3. Mental retardation is present in many diseases.
4. The most common diseases are **Gaucher's** disease and **Niemann Pick** diseases (see the table below).

Summary of the important lipidosis

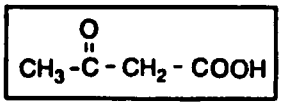
Disease	Enzyme Deficiency and Reaction Involved	Clinical Symptoms
Gaucher's disease	Glucocerebrosidase	Enlarged liver and spleen, mental retardation
Fabry's disease	Ceramide trihexosidase	Skin rashs, kidney failure ; full symptoms only in males (X-linked recessive)
Metachromatic leukodystrophy	Sulfatidase	Progressive nervous disorders due to demyelination; motor dysfunction
Krabbe's disease	Galactocerebrosidase	Severe mental retardation in infants; myelin almost absent
Niemann-Pick disease	Sphingomyelinase	Enlarged liver and spleen due to accumulation of sphingomyelin; fatal in early life
Tay-Sachs disease	Hexosaminidase	Mental retardation, blindness, demyelination, accumulation of Tay-Sachs ganglioside (G_{M2})
Generalized (G_{M1}) gangliosidosis	Galactosidase	Mental retardation, liver enlargement, skeletal deformation, accumulation of G_{M1} gangliosides

NOTE:
1- Defects in synthesis of phospholipids and sphingolipids ➔ Multiple sclerosis.
2- Defects in degradation of phospholipids and sphingolipids (lysosomal disorders) ➔ Sphingolipidosis.

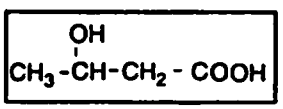
Acetone (Ketone) Bodies

I. ***DEFINITION:*** These are 3 compounds formed by the liver and include:

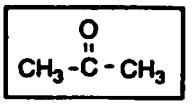
Acetoacetate



β-Hydroxybutyrate



Acetone



II. *FUNCTIONS (IMPORTANCE) OF KETONE BODIES:*

A. **Source of energy**: ketone bodies are used as a source of energy. They are converted into acetyl-CoA which is oxidized in tricarboxylic acid (TCA) cycle.

B. Skeletal muscles, cardiac muscles, kidneys and most of tissues can use ketone bodies as a source of energy in prolonged fasting and starvation.

C. **Brain tissue** can also oxidize ketone bodies within 5 to 6 days of starvation. Note: brain never oxidizes fatty acids.

D. **Liver does not** contain enzymes for ketone bodies oxidation (ketolysis). Thus liver cannot oxidize them.

III. *SYNTHESIS OF KETONE BODIES (KETOGENESIS):*

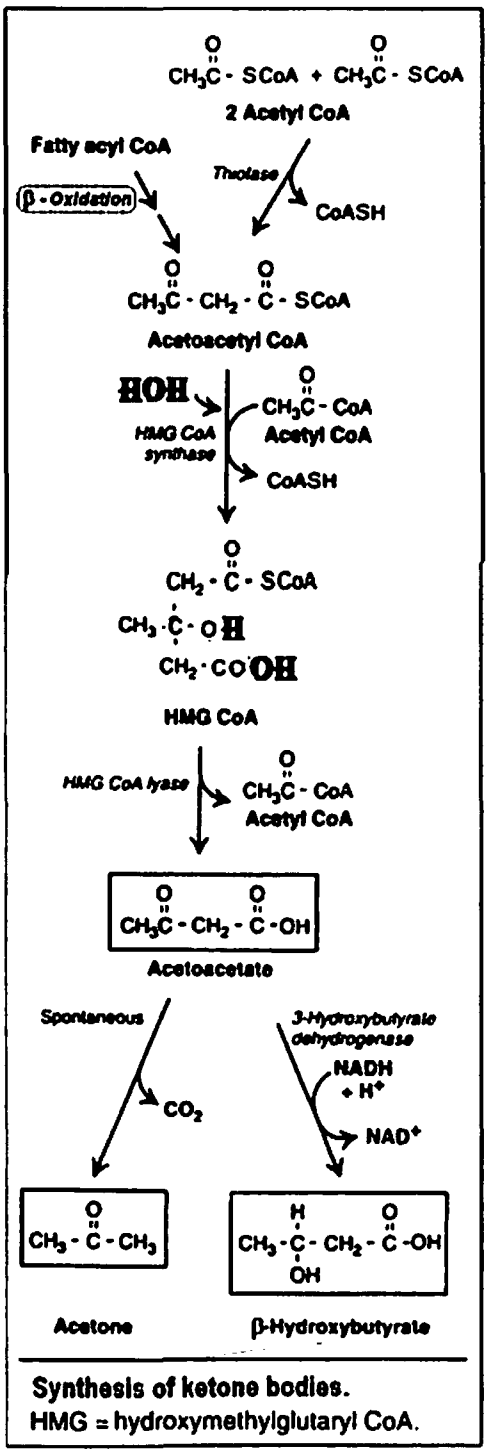
A. **Site:**

1. **Organ location:** Liver
2. **Intracellular location:** Mitochondria.

B. **Precursor:** Acetyl CoA (derived from fatty acids oxidation and ketogenic amino acids).

C. **Steps:** Acetoacetate is the first ketone body produced. Then both β-hydroxy butyrate and acetone are derived from it :

1. **Formation of acetoacetyl CoA:**
   a) From condensation of 2 acetyl CoA molecules.
   b) Acetoacetyl CoA may also be derived from β-oxidation of fatty acids (last 4 carbons).
2. **Formation of HMG-CoA (β-hydroxyl β-methyl glutaryl CoA):** By condensation of third molecule of acetyl CoA in the presence of HMG CoA synthase.
3. **Formation of acetoacetate** by HMG CoA lyase.



Synthesis of ketone bodies.
HMG = hydroxymethylglutaryl CoA.

4. **Formation of β- hydroxybutyrate and acetone:** Acetoacetate is either:
   a) Spontaneously decarboxylated into acetone.
   b) Reduced by hydroxybutyrate dehydroganase into β-hydroxy-butyrate.

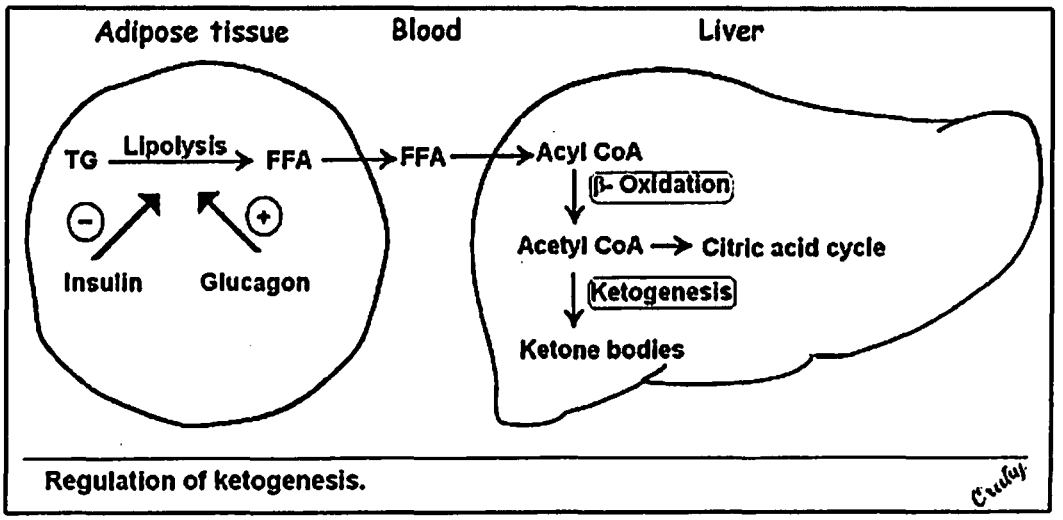
D. Conditions that increase ketogenesis:

1. Starvation.
2. Diabetes mellitus.
3. Low carbohydrate diet.
4. Hhypercatabolic states e.g. fevers, hyperthyroidism.

Note that both lipolysis and ketogenesis are activated by the same conditions.

E. Regulation of ketogenesis:

1. **After meal,** insulin is secreted, **and this inhibits lipolysis and in turn inhibits ketogenesis.**
2. **During fasting** anti-insulin hormones are secreted e.g. glucagon. **This stimulates lipolysis and in turn ketogenesis.**
3. **Explanation:**
   a) Conditions causing excessive lipolysis in adipose tissue leads to very high level of plasma free fatty acids (FFA).
   b) The liver in both fed and fasting states has the ability to **extract 30%** of the free fatty acids passing through it. Thus at high concentration of plasma FFA, the amount extracted by the liver is increased.
   c) In liver, FFA is activated to acyl CoA ➡ β-Oxidation to give acetyl CoA.
   d) Acetyl CoA is oxidized in citric acid cycle, but if the amount of acetyl CoA molecules is more than the capacity of CAC, they are used to form ketone bodies.



Regulation of ketogenesis.

IV. *OXIDATION OF KETONE BODIES (KETOLYSIS):*

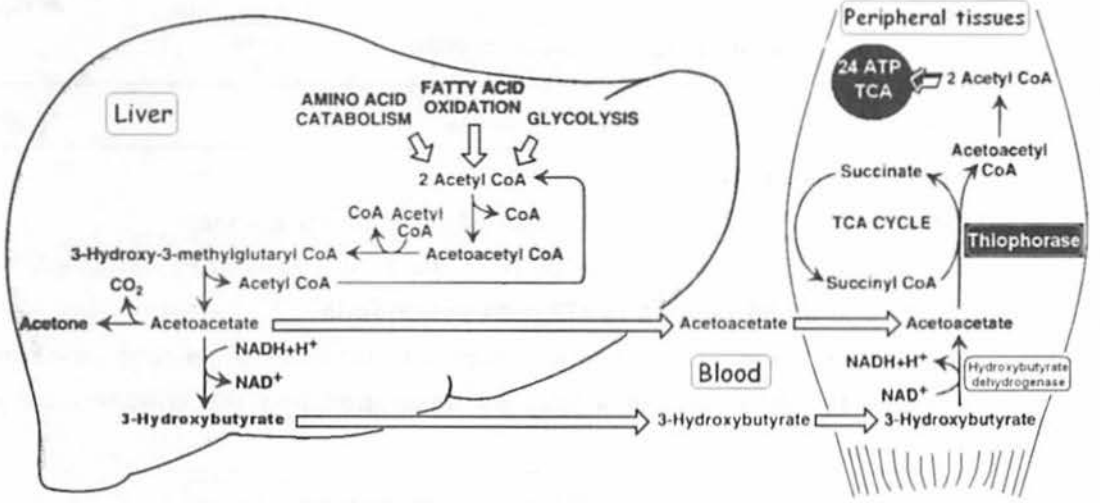
A. Site:

1. **Intracellular location:** Mitochondria.
2. **Organ location:** Extra hepatic tissues as muscles.

   Note: Ketolysis does not occur in liver because it does not contain enzymes for it.

B. Steps:

1. Acetone is volatile and removed in the expired air.
2. β-Hydroxybutyrate is converted into acetoacetate by hydroxybutyrate dehydrogenase enzyme.
3. Acetoacetate is then activated into acetoacetyl CoA by:
   a) **Thiophorase enzyme** in the presence of succinyl CoA.
   b) Acetoacetyl CoA synthetase enzyme, in the presence of ATP.
4. Acetoacetyl CoA is split into two molecules of acetyl CoA which are oxidized in citric acid cycle. Each gives 12 ATP.



Ketone bodies synthesis in the liver [KETOGENESIS] and utilization in peripheral tissues [KETOLYSIS].

V. *BLOOD KETONE BODIES AND KETOSIS:*

A. Blood ketone bodies concentration is less than 3 mg/dl.

B. Ketonemia: is the increase of blood ketone bodies above normal concentration.

1. It occurs when the rate of formation of ketone bodies (ketogenesis) is grater than the rate of their oxidation (ketolysis).
2. If the condition is severe, ketonemia may lead to acidosis (**ketosis**).

C. Urine ketone bodies: is less than 40 mg/day.

D. Ketonuria: is the increase of urine ketone bodies concentration above normal concentration:

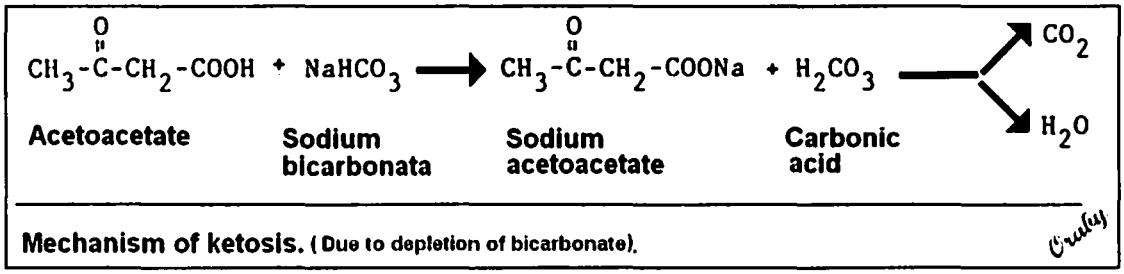
1. It usually occurs with ketonemia.

E. Causes of ketonemia and ketonuria:

1. Starvation
2. Severe diabetes mellitus.
3. Hypercatabolic states e.g. diarrhea and fever.

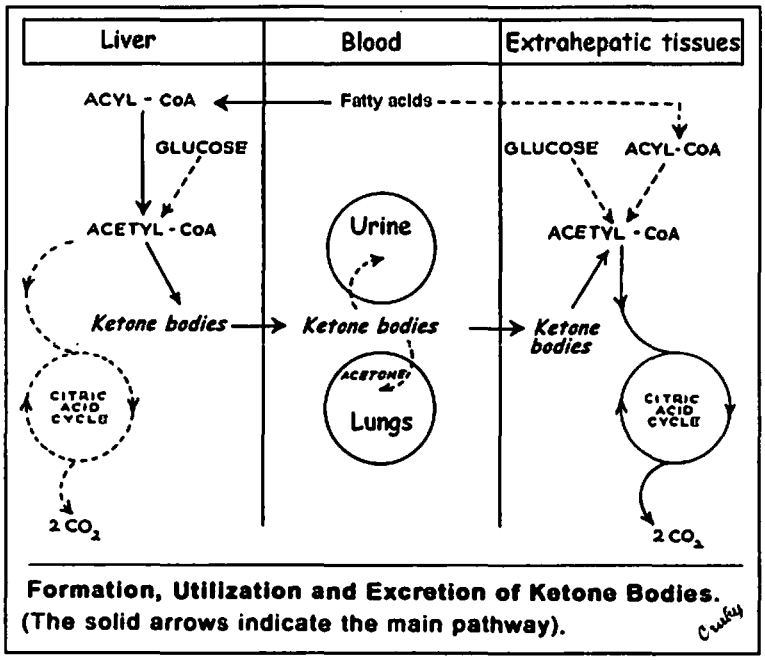
F. Ketosis (= ketoacidosis):

1. **Definition**: It is a condition of **metabolic acidosis** results from ketonemia.
2. **Mechanism:**
   a) An increase of ketone bodies in the blood is neutralized by blood buffers mainly bicarbonate (HCO_3^-).
   b) Bicarbonate will be depleted and this leads to decreased blood pH (acidosis).



Mechanism of ketosis. (Due to depletion of bicarbonate).

3. **Effects of acidosis:**
   a) Acidosis causes dizziness, loss of concentration...etc.
   b) Acidosis causes transfer of potassium (K^+) ions from intracellular fluid to blood leading to **($\uparrow K^+$) hyperkalemia.**
   c) **Ketotic coma:** In severe cases of ketosis as in uncontrolled diabetes mellitus, coma may be developed and the condition may be fatal.



Formation, Utilization and Excretion of Ketone Bodies.
(The solid arrows indicate the main pathway).